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Multispectral Remote Sensing Classification using Differentiable Weightless Neural Networks

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Abstract— Remote sensing image classification plays a critical role in environmental monitoring, land-use analysis, and urban planning. Although deep convolutional neural networks achieve high classification accuracy, they often require substantial computational and memory resources, limiting deployment in edge and resource-constrained systems. This paper presents a computationally efficient framework for multispectral satellite image classification using Differentiable Weightless Neural Networks (DWNs) and handcrafted feature representations. The proposed approach combines spectral indices, band statistics, HSV histograms, and texture descriptors within a multi-scale spatial pyramid framework. The extracted features are binarized using distributive thermometer encoding and classified through lookup-table-based DWN inference. Ablation studies and hyperparameter analysis evaluate the contribution of individual feature groups and LUT configurations. Experimental results on the EuroSAT benchmark show that the selected configuration (L=8 thermometer levels and n=4 LUT inputs) achieves 97.00% ± 0.05% (Cohen's κ = 0.9666, macro-F1 = 0.9686) on the full 13-band multispectral variant and 93.24% ± 0.12% on the RGB variant, with a total deployable footprint of 0.091 MB — including LUT contents, the learnable input mapping, and the thermometer encoder — over 492× smaller than ResNet-18 and roughly 100× smaller than MobileNetV2. The improvement from RGB to 13-band input is consistent with the added discriminative value of red-edge, NIR, and short-wave infrared channels, although the two pipelines also differ in feature composition. The proposed method provides an interpretable and hardware-efficient alternative for remote sensing classification in resource-constrained environments.

***Keywords*—Weightless neural networks, satellite image classification, Differentiable weightless neural networks, feature extraction**

I. INTRODUCTION

Remote sensing image classification has become an important research area due to the rapid growth of Earth observation systems and the increasing availability of high-resolution satellite imagery. Multi-spectral data acquired from sensors such as Sentinel-2 provide valuable information for environmental monitoring, urban planning, and agricultural analysis and land-use assessment [1].

Traditional remote sensing classification methods primarily relied on handcrafted spectral and texture descriptors combined with machine learning algorithms such as Support Vector Machines (SVMs), Random Forests (RFs), and k-Nearest Neighbors (k-NN) [2], [3]. Spectral indices, including the Normalized Difference Vegetation Index (NDVI) and Soil Adjusted Vegetation Index (SAVI) have been widely used for vegetation and land-cover analysis [4], [5]. In addition, texture descriptors such as Gray-Level Co-occurrence Matrix (GLCM), Local Binary Patterns (LBP), and Histogram of Oriented Gradients (HOG) have shown strong effectiveness in capturing spatial information from satellite imagery [6]–[8].

In recent years, deep convolutional neural networks (CNNs) have achieved state-of-the-art performance in remote sensing image classification tasks [9], [10]. Architectures such as ResNet, DenseNet, and EfficientNet automatically learn hierarchical feature representations directly from raw imagery and significantly improve classification accuracy [10]–[12]. However, these models often require large training datasets, extensive computational resources, and long training times. Furthermore, deep learning systems are frequently criticized for limited interpretability and high model complexity.

Weightless Neural Networks (WNNs), particularly Differentiable Weightless Neural Networks (DWNs) [14], offer a promising alternative for efficient learning. Unlike conventional neural networks that rely heavily on arithmetic operations, WNNs utilize a memory-based lookup table (LUT) for inference, resulting in faster execution and lower energy consumption [13]. Since WNNs operate on binary inputs, continuous-valued features are first converted using Distributive Thermometer Encoding [21]. DWNs extend WNNs with differentiable, gradient-based training, making them well-suited for resource-constrained and real-time applications.

Motivated by these advantages, this work proposes a lightweight framework for multispectral remote sensing image classification using Differentiable Weightless Neural Networks. The proposed approach combines handcrafted spectral, statistical, histogram, and texture descriptors with Distributive Thermometer Encoding and differentiable

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lookup-table classification. Unlike conventional CNN-based approaches that rely entirely on deep feature learning, the proposed framework explicitly integrates physically meaningful spectral information and interpretable texture representations.

The main contributions of this paper are summarized as follows:

1. A comprehensive handcrafted feature extraction framework for EuroSAT RGB and 13-band multispectral imagery.
2. Integration of spectral indices, histogram representations, statistical descriptors, and texture features into a unified DWN-based classification pipeline.
3. A leave-one-out feature-group ablation study over five random seeds, reported with accuracy, Cohen's kappa, macro-F1, and a per-class confusion analysis.
4. A hyperparameter sweep over thermometer encoding levels and LUT address sizes, with model selection performed strictly on a held-out validation split and final test evaluation of the locked configuration over five seeds.
5. Directly measured baselines on the identical data split: six 13-band CNN architectures trained by the authors, and four conventional classifiers trained on the identical 860-dimensional feature vector, isolating the contribution of the DWN beyond the features.
6. A total deployable model footprint of 0.091 MB — including LUT contents, the learnable input mapping, and the thermometer encoder — over 492× smaller than ResNet-18, demonstrating that DWN-based inference with domain-informed features approaches CNN accuracy at a fraction of the model complexity.

The paper is organized as follows: Section II reviews related work. Section III details the methodology. Section IV presents experimental results and discussion. Section V concludes the paper.

II. Related Work

A. Dataset

The EuroSAT dataset is widely used for land use and land cover classification and contains 27,000 labeled images distributed across ten classes [1]: Annual Crop, Forest, Herbaceous Vegetation, Highway, Industrial, Pasture, Permanent Crop, Residential, River, and SeaLake. We use the standard 80:20 stratified train/test split, evaluating both the RGB and full 13-band multispectral variants. The multispectral version includes red-edge, NIR, water-vapor, cirrus, and SWIR bands unavailable in RGB imagery.

B. Remote Sensing Image Classification

Remote sensing image classification has evolved from handcrafted feature descriptors with classical classifiers such as SVMs and Random Forests [2], [3] to deep CNNs, including ResNet [10], DenseNet [11], and EfficientNet [12], which achieve excellent accuracy by learning hierarchical feature representations. However, these models require substantial computational resources and large parameter counts, motivating lightweight alternatives for edge deployment. Despite their success, CNN-based methods often require extensive computational resources and large-scale labeled datasets. Consequently, lightweight and interpretable alternatives remain an important area of research.

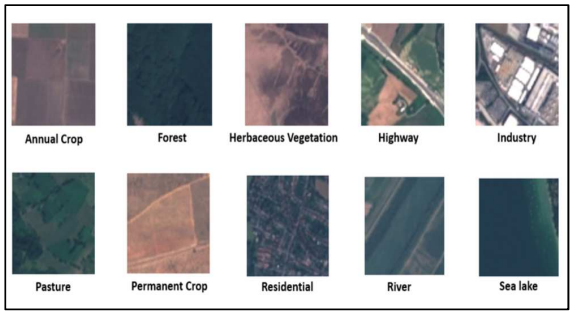


Fig. 1. EuroSAT dataset sample images across 10 LULC classes

C. Feature Descriptors

Handcrafted descriptors remain attractive for remote sensing because they are interpretable and computationally efficient. Spectral indices (e.g., NDVI, NDWI, SAVI), texture descriptors (GLCM and LBP), and gradient features (HOG) capture complementary spectral, structural, and spatial information, improving land-cover discrimination [4]–[8].

D. Lightweight models and Edge deployment

Deploying deep learning models on resource-constrained hardware has motivated lightweight architectures, model compression, and FPGA accelerators [18]–[20]. While MobileNetV2 and EfficientNet substantially reduce model size compared with conventional CNNs, they still require millions of parameters and floating-point computation. In contrast, DWNs replace arithmetic operations with LUT-based inference, enabling extremely compact and hardware-efficient classifiers [14].

E. Weightless Neural Networks: From WiSARD to DWN

WNNs store learned information in LUTs rather than weighted connections, enabling inference through simple memory accesses instead of multiply–accumulate operations, which results in fast, energy-efficient, and hardware-friendly computation [13]. Early WNN architectures, such as WiSARD, relied on fixed input mappings and non-differentiable training procedures. DWNs extend this paradigm by introducing Extended Finite Difference–based optimization together with learnable input mappings and learnable reduction layers, enabling end-to-end gradient-based training while preserving LUT-based inference [14]. These advances significantly improve the representational capacity and adaptability of WNNs, making them suitable for more complex vision tasks.

Although handcrafted descriptors and DWNs have each been studied independently, their integration for multispectral remote sensing classification has received little attention. This work combines multi-scale handcrafted descriptors with DWN inference to achieve accurate and

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compact land-cover classification suitable for edge deployment.

III. Methodology

A. Proposed Model

The proposed framework, illustrated in Fig. 2, consists of four stages: multi-scale descriptor extraction, feature fusion and binarization, LUT-based DWN inference, and GroupSum classification. Features are independently extracted from the full 64×64 image and a central 32×32 crop to capture both global context and local spatial structure. The resulting descriptors are concatenated to form a fixed-length feature vector and binarized using Distributive Thermometer Encoding with 8 levels. This binary vector is processed by a DWN followed by GroupSum, which pools activations into 10 class scores for final classification.

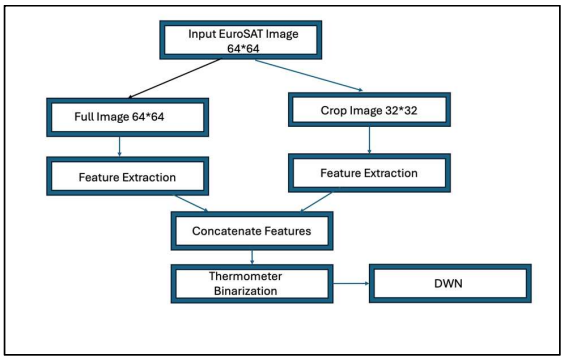


Fig. 2. Proposed DWN-based multispectral classification pipeline including multi-scale feature extraction, thermometer encoding, LUT inference, and GroupSum classification.

B. Feature Extraction Pipeline

A comprehensive feature extraction pipeline is developed to capture spectral, spatial, and statistical characteristics of the multispectral imagery. The extracted features, listed below, provide a rich representation of the underlying data and enhance the model's ability to accurately distinguish between different land-cover classes and scene characteristics.

1) Spectral Indices: Six vegetation and water indices (NDVI, NDWI, NDBI, EVI, SAVI, RE-NDVI) are computed per pixel. A small epsilon (10^{-6}) prevents division by zero. The mean, standard deviation, skewness, and kurtosis of each index map are extracted.

2) Band statistics: For each of the 13 spectral bands, we compute the mean, standard deviation, skewness, kurtosis, 25th percentile, and 75th percentile. These statistics capture global spectral distributions and improve class separability.

3) HSV Histograms: The RGB composite from bands B02, B03, and B04 is converted to HSV. An 8-bin normalized histogram is extracted per channel, yielding 24 features per scale.

4) Band Histograms: An 8-bin intensity histogram over [0,1] is computed for each of the 13 bands.

5) Texture features: They are extracted to capture spatial patterns and structural information that cannot be represented using spectral information alone. In remote sensing imagery, different land-cover classes often exhibit distinctive texture characteristics, making texture analysis important for accurate classification.

In this work, multiple complementary texture descriptors are employed. GLCM properties are retained per angle rather than averaged across angles: contrast, dissimilarity, homogeneity, and energy are extracted from the grayscale GLCM at each of the two angles (8 values), and contrast from the NIR (B08) GLCM at each angle (2 values), giving 10 texture-matrix features per scale. GLCM features are used to model spatial relationships between neighboring pixels through measures such as contrast, homogeneity, dissimilarity, and energy.

Uniform LBP with P=8 neighbors and radius R=1 is computed on the luminance channel. A 10-bin histogram is extracted per scale. LBP is utilized to capture local texture variations by encoding neighborhood intensity relationships into binary patterns. In addition, HOG is computed on the luminance channel with 9 orientations, 16×16-pixel cells, and 2×2-cell blocks. HOG descriptors are extracted to represent edge orientation and structural information within the satellite imagery. The combination of these texture descriptors enhances the discriminative capability of the proposed framework, particularly for visually similar classes such as residential, industrial, highway, and agricultural regions.

The luminance channel used for LBP, HOG, and the grayscale GLCM is computed as

$$L = 0.2989 * \mathrm{R} + 0.5870 * \mathrm{G} + 0.1140 * \mathrm{B} \quad (1)$$

where $\mathrm{R} = \mathrm{B04}, \mathrm{G} = \mathrm{B03}$ and $\mathrm{B} = \mathrm{B02}$.

The same band ordering is used for the HSV conversion.

C. Spatial Pyramid Pooling

All features are extracted twice: once on the full 64×64 image and once on the 32×32 center crop. This two-scale spatial pyramid captures both global land-cover context and local spatial texture.

D. Thermometer Encoding and DWN inference

Real-valued features are binarized using thermometer encoding. For each feature dimension, a set of 8 thresholds is learnt from the training distribution, and each threshold contributes one bit to the encoded representation. This expands the 860-dimensional real vector to an exactly 6,880 (860 × L-dimensional binary vector) bits at the selected L=8.

For each feature dimension f_i with L fitted thresholds $\tau_1<\tau_2<\cdots<\tau_L$

$$b_k = \begin{cases} 1, & f_i \geq \tau_k \\ 0, & otherwise \end{cases} \quad (2)$$

where τ_k is the threshold for level k, where k = 1, 2, …, L. The thresholds $\tau_1 < \tau_2 < \cdots < \tau_L$ are not fixed uniformly but are fitted to the training data distribution using the Distributive Thermometer approach [21], which ensures that each threshold level captures a meaningful quantile of the feature space, producing a more discriminative binary representation than uniform quantization. The DWN processes the binary vector through an LUT layer whose pooled group activation yields the class prediction.

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IV. Experiments And Results

A. Evaluation Protocol and DWN model configuration

The EuroSAT dataset was divided into stratified training (80%) and test (20%) sets using a fixed random seed (42). To select the DWN configuration, 10% of the training data was held out as a validation set. A grid search over different thermometer levels and LUT address sizes was performed on the validation set using three random seeds. The best configuration (L=8, n=4) was then retrained on the full training set and evaluated once on the untouched test set over five random seeds.

The DWN was implemented using the torch_dwn library [14] with an 8,000-node LUT layer followed by a GroupSum classifier. Each node receives four binary inputs through a learnable mapping and stores a 16-entry Boolean lookup table. Models were trained using Adam with cross-entropy loss, an initial learning rate of 10^{-2}, StepLR decay every 30 epochs, a batch size of 100, and 100 training epochs.

Each LUT node stores $2^4 = 16$ one-bit entries, giving a total LUT memory of 8,000 × 16 × 1 bit = 128 Kbits. In a CPU or microcontroller deployment, the learned input mapping must also be stored: each node records which 4 of the 6,880 input bits it reads, requiring 8,000 × 4 × $\lceil \log_2 6{,}880 \rceil$ = 416 Kbits. The thermometer encoder stores 860 × 8 = 6,880 learned thresholds, requiring 220.2 Kbits using 32-bit floating-point representation (110.1 Kbits using FP16). The resulting deployable model occupies 0.091 MB, including LUT contents, learnable mapping, and thermometer thresholds. On FPGA, the input mapping is realized as routing rather than stored memory, reducing on-chip storage to 348.2 Kbits (0.042 MB) while requiring no DSP blocks. residing entirely within BRAM with zero DSP utilization; for conservatism, all size comparisons in this paper use the full 0.091 MB figure. Inference consists only of threshold comparisons, LUT accesses, and integer accumulation, requiring no multiply–accumulate operations.

B. Comparison with CNN-Based Architectures

Table I compares the proposed DWN with CNN baselines trained by the authors under an identical protocol. The proposed DWN achieves 97.00% ± 0.05% on the EuroSAT 13-band model, matching ShuffleNetV2 (97.00%), within 0.1 percentage points of ResNet-18 (97.07%), and outperforming SqueezeNet (95.98%), while remaining 1.4–1.5 points below MobileNetV2 and EfficientNet-B0. Its primary advantage is deployment efficiency rather than peak accuracy: the complete deployable model occupies 0.091 MB, approximately 15× smaller than the smallest float32 CNN, 100× smaller than MobileNetV2, and 492× smaller than ResNet-18, while requiring only threshold comparisons, LUT accesses, and integer accumulation. It is important to note the paradigm trade-off in this comparison: CNNs perform feature learning within their parameters, whereas the DWN relies on a pre-computed feature-extraction stage whose computational cost is analyzed later in this paper.

The RGB pipeline uses the same feature extractor and DWN configuration as the 13-band model, excluding descriptors requiring unavailable spectral bands (spectral indices, per-band statistics, per-band histograms, and NIR GLCM). The remaining descriptors comprise ExG statistics, RGB statistics, HSV histograms, grayscale GLCM, LBP, and HOG, yielding 456 features (3,648 binary bits) and a deployable footprint of 0.075 MB. Using the operating point selected on the 13-band validation set (L=8, n=4), the RGB pipeline achieves 93.24% ± 0.12%. The 3.76-point gap relative to the multispectral model reflects both the additional discriminative information provided by the red-edge, NIR, and SWIR bands and the reduced feature representation.

TABLE I. Performance comparison of different models on the EUROSAT dataset

Model	Accuracy	Model Size
EuroSAT RGB Variant		
VGG-16 [1],[15]	95.0 – 97.5	~552 MB
ResNet-50[1],[10]	97.0 – 99.0	~102 MB
ResNet-18[1],[10]	96.0 – 98.5	~47 MB
MobileNetV2[1],[16]	94.0 – 97.0	~14 MB
EfficientNet-B0[1],[12]	94.5–97.5	~21 MB
ViT-B/16[1],[17]	96.0 – 98.0	~344 MB
SVM + HOG [1]	85.0 – 89.0	~0.4 MB
Proposed DWN	93.24 ± 0.12	0.075 MB
EuroSAT 13-band Multispectral Variant		
ResNet-18	97.07	44.8 MB
Shufflenet V2 ×0.5	97.00	1.40 MB
EfficientNet-B0	98.46	16.1 MB
MobileNetV3-Small	97.50	6.12 MB
MobileNetV2	98.37	8.96 MB
SqueezeNet 1.1	95.98	2.92 MB
Proposed DWN	97.00 ± 0.05	0.091 MB

**13-band CNN baselines are measured by the authors on the identical stratified split (seed 42) with the first convolution adapted from 3 to 13 input channels (RGB-mean initialization); checkpoints are selected by validation accuracy on the same 2,160-image validation split used for DWN selection, and the test set is evaluated once after training. CNN sizes = parameters × 4 bytes (float32), excluding activations. The DWN size includes LUT contents, learnable input mapping, and thermometer encoder; feature extraction is performed as preprocessing, and its code size is excluded (CNN parameter counts, by contrast, include feature-learning capacity).*

Table II presents the hyperparameter sweep over thermometer encoding level L ∈ {4, 6, 8, 10} and LUT address size n ∈ {4, 6, 8}, evaluated over three random seeds. Validation accuracy is remarkably insensitive to (L, n) within the grid: all twelve configurations lie between 96.70% and 97.15%, and differences between neighboring configurations are largely within one standard deviation, indicating that the handcrafted feature representation, rather than the precise LUT geometry, determines performance. We therefore select the argmax of mean validation accuracy, L=8 and n=4 (97.15% ± 0.12%), and note that this configuration is simultaneously the most hardware-economical in the grid: 4-input nodes store only 16 LUT entries each, and the resulting deployable footprint of 0.091 MB is the smallest among all twelve configurations. A 4-input Boolean function also fits trivially within a single FPGA LUT6 primitive (fracturable LUT6 architectures can pack two such functions per primitive), preserving a direct node-to-logic mapping with zero DSP utilization. We do not draw trend conclusions about L or n from Table II, as the observed differences are within seed noise. The locked

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configuration, retrained on the full training pool, achieves 97.00% ± 0.05% test accuracy over five seeds.

TABLE II. HYPERPARAMETER SWEEP OF EUROSAT MULTISPECTRAL DATASET: VALIDATION ACCURACY (%) VS. THERMOMETER LEVELS (L) AND LUT ADDRESS SIZE (N)

Validation Accuracy (%)				
L	**Feature Size**	**LUT Address size**		
		4	6	8
4	3440	97.02 ± 0.19	97.01 ± 0.09	96.77 ± 0.09
6	5160	96.74 ± 0.13	96.84 ±0.06	96.88 ± 0.02
8	6880	**97.15 ± 0.12**	97.01± 0.06	96.99 ± 0.04
10	8600	96.87 ± 0.04	96.91± 0.06	96.70 ± 0.06

TABLE III. FEATURE ABLATION STUDY FOR EUROSAT 13- BAND MULTISPECTRAL DATASET for L=8, n=4.

Feature Extractor	Test Accuracy (%)	Cohen's κ	macro-F1
Full (all features)	97.00 ± 0.05	0.9666±0.0006	0.9686 ± 0.0006
without texture	96.11 ± 0.12	0.9568 ± 0.0013	0.9594 ± 0.001
without Indices	96.57 ± 0.06	0.9618 ± 0.0007	0.9642±0.0006
without crop pyramid	96.79 ± 0.07	0.9642 ± 0.0008	0.9661 ± 0.0008

Table III reports a leave-one-group-out ablation at the selected operating point, retraining each configuration from scratch over five seeds. Removing texture descriptors (GLCM, LBP, HOG; 400 of 860 dimensions) causes the largest drop of 0.89 points, confirming that spatial structure remains the most discriminative signal — consistent with classes such as Residential, Industrial, and Highway sharing similar spectral profiles but differing in spatial arrangement. Removing spectral indices costs 0.43 points, reflecting the value of physically meaningful reflectance ratios for separating vegetation, water, and built-up classes, and removing the crop pyramid costs 0.21 points. Notably, the texture contribution is concentrated in the compact GLCM and LBP descriptors: removing HOG alone (360 dimensions, 42% of the feature vector) leaves accuracy statistically unchanged, indicating that a leaner 500-feature variant without HOG would substantially shrink both the encoder and the input mapping at no accuracy cost — a direction left to future work. No removal improves accuracy, validating the multi-modal design; Cohen's kappa and macro-F1 track accuracy across all configurations.

Per-class analysis of the final model in Fig. 3 shows near-perfect recognition of spectrally distinctive classes — Forest (99.2% recall), SeaLake (99.3%), Residential (98.8%) — while Highway is the weakest class (89.8% recall, F1 = 0.918), most frequently confused with Industrial (11 cases), AnnualCrop (9), and PermanentCrop and River (8 each): narrow linear structures at 10 m ground resolution share both spectral and textural characteristics with adjacent land cover. PermanentCrop and AnnualCrop show the expected mutual confusion among agricultural classes.

To isolate the DWN's contribution beyond the feature pipeline, conventional classifiers are trained on the identical 860-dimensional feature vector and split. The DWN outperforms all of them, including a size-matched MLP by 1.6–1.7 points and gradient boosting by 0.37 points at 45× less storage, while requiring no floating-point arithmetic at inference as shown in Table IV. The features alone therefore do not explain the reported accuracy; the LUT-based classifier contributes measurably.

To assess the contribution of feature engineering, the DWN was evaluated using handcrafted features and raw flattened pixels under an identical reduced configuration (4,000 LUT nodes, L=6, n=6, 32×32 input) as summarized in Table V. The reduced configuration was necessary to fit the raw-pixel learnable mapping in GPU memory. Using handcrafted features, the DWN achieved 96.22%, compared with 85.78% using raw pixels, a 10.44-point improvement despite the raw representation being approximately 25× larger (13,312 vs. 536 features; 79,872 vs. 3,216 binary bits). These results demonstrate that the proposed feature representation provides a compact, information-rich descriptor space that improves both classification accuracy and computational efficiency.

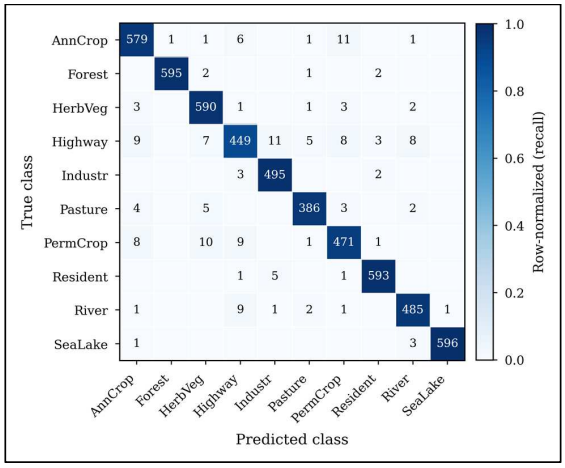


Fig. 3. Confusion Matrix of the final model, from the seed whose accuracy is closest to the five-seed mean.

TABLE IV. CLASSIFIER BASELINES ON IDENTICAL FEATURES

Classifier	Test Accuracy (%)	Size
Logistic regression	94.63	34 KB
MLP (1 hidden layer, width 100)	95.29	0.17 MB (fp16)
MLP (1 hidden layer, width 100)	95.44	0.087 MB (int8)
Histogram gradient boosting	96.63	4.1 MB
DWN (L=8, n=4)	97.00	0.091 MB

On a single CPU core, feature extraction requires 62.7 ± 0.5 ms per image (batch size 1), dominated by GLCM computation. Thermometer encoding adds 0.02 ms, while the DWN forward pass requires 5.8 ± 0.8 ms using the *torch_dwn* GPU implementation (the library provides no CPU implementation). The end-to-end pipeline therefore requires 68.5 ms per image, with over 90% of the latency

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attributable to feature extraction rather than LUT-based inference. The DWN classifier performs only threshold comparisons, LUT accesses, and integer accumulation, requiring no multiply–accumulate operations. Future work will accelerate the descriptor stage through GPU/FPGA implementations and lightweight binary-native alternatives to computationally intensive descriptors such as GLCM and HOG. The intended FPGA implementation stores the complete classifier in BRAM with zero DSP utilization.

TABLE V. TEST ACCURACY OF DIFFERENT CONFIGURATIONS WITH AND WITHOUT FEATURE EXTRACTION (FE)

Configuration	Feature Dimension	Binarized Dimension	Test Accuracy
With FE 64×64, L=8	860	6880	97.00 ± 0.05
With FE 32×32, L=6	536	3,216	96.22
Without FE 32×32, L=6	13,312	79,872	85.78

**Single training run at the reduced configuration (4,000 LUT nodes, L=6, n=6, 32×32 input), required to fit the raw-pixel learnable mapping in GPU memory; both reduced rows use identical training settings, making the comparison internally matched.*

V. CONCLUSION

This paper presents a hardware-efficient multispectral LULC classification framework combining a handcrafted 860-dimensional spatial-pyramid feature pipeline with Differentiable Weightless Neural Network inference. Under a strict validation-based selection protocol, hyperparameters chosen on a held-out validation split, the selected configuration (L=8 thermometer levels, n=4 LUT inputs) achieves 97.00% ± 0.05% accuracy on the 13-band EuroSAT variant and 93.24% ± 0.12% on the RGB variant. The complete deployable model occupies 0.091 MB — over 492× smaller than ResNet-18 — with inference requiring no multiply-accumulate operations. On an identical train/validation/test split, the proposed DWN matches ShuffleNetV2, approaches ResNet-18 in accuracy while using 15–492× less storage, and outperforms all conventional classifiers trained on the same feature representation, including gradient boosting, demonstrating the effectiveness of LUT-based classification beyond handcrafted features alone. Feature ablation over five seeds identifies texture as the most critical group, followed by spectral indices and the crop pyramid. End-to-end latency is dominated by classical feature extraction rather than the classifier; accelerating the descriptor stage or developing more computationally efficient descriptors, and FPGA synthesis of the complete BRAM-resident pipeline, are left to future work.

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